

# Rabin Shrestha

📍 Kathmandu, Nepal · Remote — US & EU overlap hours

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Senior Software Engineer · Fintech · AI & LLM Applications

## SUMMARY

Software engineer with 12+ years of backend and distributed-systems experience, now building **LLM-backed products end to end**. Seven of those years in **fintech** as **founding engineer** at ZenLedger — grew it from startup stage through **Series B**, led a team of up to 15, owned the financial data platform behind **10B+ transaction rows** and **\$50B+ in tracked assets**, and ran engineering on its **crypto-forensics product used by the IRS**.

## EXPERIENCE

### Marpha Labs — Co-Founder & Engineering Lead

Remote · 2025 – Present

*An independent, self-funded product studio — a distributed collective of builders across Nepal, the USA, Canada, and Australia.*

#### SwiftScout LIVE ON APP STORE

Native iOS · Swift

- Shipped a native iOS app alerting drivers to **50+ enforcement cameras across San Francisco and Oakland**, processed **entirely on-device** for privacy — live on the App Store as a **paid subscription** product.

#### FillAnyForms PRE-LAUNCH

LLM infra · Rails · AWS

- Built an **AI-assisted document processing system** that extracts fillable fields from PDFs — and from scans via image-to-text — and maps structured user data into them.
- Built the LLM layer end to end: a **LiteLLM proxy** over OpenRouter giving multi-provider model routing, virtual API keys per consumer, and **per-user usage and cost tracking** with per-account spend caps.
- Added a **privacy mode** routing inference to the user's own Ollama or OpenAI-compatible endpoint, so document data never leaves their machine.

#### HabreCare IN DEVELOPMENT

React Native · Expo · Supabase

- Built a mental-wellbeing app for anxiety, insomnia, and overthinking in React Native and TypeScript, for iOS and Android.
- Developed **spec-first with AI coding agents**: design specs and a decision log per feature, with automated typecheck, lint, and test gates every agent-authored change must pass before merge.
- Architected **local-first sync** — writes land on-device instantly and offline, then reconcile to Supabase under per-user row-level security.

### ZenLedger.io — Founding Engineer & Lead Software Engineer

Remote · 2018 – 2025

**Fintech** platform for tax, accounting, and regulatory reporting on crypto assets, with a **forensics product line** for government. Joined at startup stage as a founding engineer and scaled the product and engineering team through its **Series B** round, on a team distributed across North America, Europe, and Nepal.

#### BATS FEDERAL CONTRACT

Engineering Lead · crypto forensics

- Engineering lead** for ZenLedger's government crypto-forensics product, used by the **IRS Criminal and Civil Investigation** units and state tax authorities across a **multi-year federal contract**; built the multi-tenant investigator platform with agency-scoped authorization and **PII segregation**.
- Built the **blockchain data platform** behind the product — replica **Bitcoin and Ethereum nodes** whose full transaction history was traversed into queryable stores, Bitcoin into **PostgreSQL** and Ethereum into **MongoDB**, serving both investigators and models.
- Built the **training-data pipeline for address attribution** — **827K labelled addresses across 29 classes**, each a **76-feature behavioural vector** derived from that index — and the service that ran the trained classifier across every block to attribute addresses at chain scale.

## ZenLedger Platform

Rails · Go · AWS

- Owned the **financial data ingestion** and parsing platform integrating **100+ exchanges and blockchains**, processing **10B+ transaction rows** and tracking **\$50B+ in customer assets** across **100K+ customers**.
- Led an engineering team of **up to 15**, setting engineering standards, owning delivery timelines, and driving architecture and specification reviews across teams.
- Served as the core engineer on the tax calculation and automated report generation pipeline, and **rewrote the calculation service from Rails/Sidekiq to Go** — **70% less compute** for the same workload, running **60% faster**, cutting compute spend proportionally.
- Built and operated the payment system, and designed **autoscaling AWS infrastructure** that matched capacity to highly seasonal demand, protecting availability at peak while holding down off-peak cost.

## EB Pearls — Software Engineer → Associate Project Manager

Kathmandu · 2013 – 2018

- Delivered **6+ enterprise projects** for an Australia-based client in PHP (CodeIgniter, Laravel) and MySQL, from requirements through release.
- Promoted into delivery ownership as Associate Project Manager — ran delivery across a **10-person** engineering, QA, and design group, owning scope changes, budgets, and release timelines.

## TECHNICAL SKILLS

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**Fintech** — Payments · financial data ingestion & reconciliation · high-volume transaction processing · tax and regulatory reporting · blockchain forensics & address attribution for government & law enforcement

**AI & ML** — Supervised classification · feature engineering · training-data pipelines · model inference at scale · LLM application design · LiteLLM · OpenRouter · multi-provider routing · virtual keys, usage & per-user cost tracking · local-inference privacy modes (Ollama, OpenAI-compatible) · image-to-text extraction · spec-driven development with AI coding agents

**Core** — Go · Ruby on Rails · TypeScript · React · PostgreSQL · Redis · AWS · Docker

**Strong** — React Native (Expo) · Next.js · Swift / native iOS · Supabase · Sidekiq · Faktory · OpenTofu/Terraform · Cloudflare (R2, CDN) · GitHub Actions · RSpec · Jest · Playwright

**Familiar** — MySQL · PHP (Laravel, CodeIgniter) · Preact · Grafana · Datadog · ScoutAPM · CircleCI · Maestro

**Leadership** — Team leadership (up to 15 engineers) · architecture & delivery ownership · Agile/Scrum

## EDUCATION

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**B.E. Computer Engineering** — Kathmandu University, Dhulikhel, Nepal

2009 – 2013

**Certified ScrumMaster (CSM)** — Scrum Alliance